Joe Baker

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Profile

I am a Software Engineer, currently working at Indra MicroNav in Maintenance, helping patch bugs and issues in their C++ Air Traffic Controller Training Simulation codebase. I Graduated from Bournemouth University last year with first class honours, following an industrial placement at MicroNav.

Previous experience includes maintaining a large legacy C++ codebase, and developing supporting tools in Python and C#. Strong foundation in object-oriented design, testing, and debugging of complex, safety-critical systems. With Uni and personal projects, also have experience across Java, Kotlin, and Web technologies, along with my current personal project involving embedded programming on an ESP32.

I am currently looking to continue developing my skills as a software engineer, contributing to meaningful projects while expanding my experience with different technologies.

Experience

Software Engineer @ Indra MicroNav (2023-Present)

Software Engineer: 2026-Present
Graduate Software Engineer: 2025-2026
Software Engineering Placement: 2023-2024

- Designed and developed an internal automated testing tool in Python to during the creation and execution of nightly regression tests with Jenkins while on the Validation team.
- Investigated and fixed bugs and errors, ranging from minor to critical, across the companies C++ Air Traffic Control simulation software suite while on the Maintenance team.
- Worked with complex, safety-critical simulation software, following established development, testing, and review processes.
- Awarded *Highly Commended* by Bournemouth University for outstanding performance and contribution during industrial placement.

Education

Bournemouth University (2021-2025)

BSc (Hons) Software Engineering. Achieving first class honours.

- Dissertation & Final Year Project (2025): using Kotlin and Python, developed a Pedestrian Navigation App which focused on the safety of provided routes, receiving a 80% grade.
- Ubiquitous Computing (2024): using React Native, developed a QR/Barcode Scanner App receiving 95% and a Surf Forecast app receiving 90%.
- Advanced Development (2024): using Python and Docker, created interconnected micro-services with others in the year group for a fictional restaurant chain, receiving 81%.
- Application Programming (2023): using Kotlin, developed a Battleships App, getting a 92% grade.
- Web Programming (2022): using PHP and JavaScript, developed an eCommerce site, with a 91% grade.
- Application of Programming Principles (2022): using Python and JavaScript, created an eBook library website with REST API, receiving a 100% grade.

Brockenhurst College (2019-2021)
BTEC Level 3 Extended Diploma in IT at grade D*D*D* (Triple Distinction Star)

Technical Skills

- **Programming Languages:** Python, C++, Java, Kotlin, HTML, JavaScript, C#, PHP, and SQL
- **Design & Architecture:** OOP (Object Oriented Programming), UML (Unified Modeling Language), REST API (Representational State Transfer API), MVC (Model View Controller), EDA (Event-Driven Architecture)
- **Platforms:** Android, Linux, Windows, and Embedded Systems
- **Software:** Docker, Git, SVN, Jenkins, MS Office, VirtualBox, and various IDEs including Android Studio, Embarcadero Builder, Visual Studio, and VS Code

Professional Skills

- **Problem Solving:** Diagnosed and resolved critical C++ software issues and bugs, improving system usability and stability.
- **Adaptability:** Quickly understood and became productive across large unfamiliar codebases, tools, and technologies, learning approaches to contribute effectively with changing roles and projects.
- **Teamwork:** Collaborated with others both at MicroNav and in university group projects, to share responsibilities and workloads, support teammates, and resolve issues, to achieve success.
- **Communication:** Presented updates and technical documentation, to both technical and non-technical audiences, while at MicroNav and during university projects.
- **Time Management:** Balanced multiple tasks and deadlines while consistently meeting deadlines and staying on schedule.

References

Available upon request